

Beatriz Carvalho Nogueira

I am a PhD Student in Informatics at Universidade de Lisboa – Faculdade de Ciências (FCUL), supported by a Mixed FCT Studentship in joint supervision with KTH Royal Institute of Technology. My research focuses on Human-Robot Interaction (HRI), Human-Computer Interaction (HCI), and accessibility, specifically investigating long-term trust calibration in HRI for older adults and the aging population. I completed my MSc in Informatics at FCUL in March 2024. My final thesis "Exploring How Robots can Help Older Adults Live by Their Values", advised by Prof. Tiago Guerreiro, received a final grade of 18/20. I also hold a Postgraduate Diploma in Statistics Applied to Biology and Health Sciences (FCUL, 2021) and a BSc in Biology from Instituto Superior de Agronomia.

My technical profile combines mixed-methods research with practical experience in data analysis and development using Python and R, foundational knowledge of Java, hands-on familiarity with robotics and design platforms (including Unity, Blender, ROS, and Aseba), and qualitative data analysis tools like Miro.

I have worked as a Full-time Researcher at LASIGE/FCUL on the European project HARIA (Human-Robot Sensorimotor Augmentation), working on an HRI Failure Taxonomy and exploring artistic expression for stroke and spinal cord injury end-users. Previously, I served as a Research Engineer in the Robotics, Perception, and Learning (RPL) division at KTH under Prof. Iolanda Leite, and as a Research Trainee at the Vasco da Gama Aquarium. My pedagogical experience includes private tutoring across multiple subjects (2016–2018). To date, I have contributed to 2 major research projects, published articles and conference contributions, including work in the ACM DIS '25 Companion and an arXiv preprint on robot failure recovery, and participated in 4 academic events and research visits (including CMU and KTH). I have also been actively involved in institutional service as a student association member and civic initiatives focused on digital transformation, climate policy, and equality.

Identification

Personal identification

Full name
Beatriz Carvalho Nogueira

Gender
Female

Birth date
1999/02/17

Citation names

Nogueira, Beatriz

Author identifiers

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Websites

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<https://beatrizcnogueira.github.io/Beatriz-Nogueira/> (Personal)

Knowledge fields

Engineering and Technology - Electrotechnical Engineering, Electronics and Informatics
Exact Sciences - Mathematics - Statistics and Probability
Natural sciences - Biological Sciences - Biology

Languages

Language	Speaking	Reading	Writing	Listening	Peer-review
English	Advanced (C1)	Advanced (C1)	Advanced (C1)	Advanced (C1)	
Portuguese (Mother tongue)					
French	Beginner (A1)	Beginner (A1)	Beginner (A1)	Beginner (A1)	Beginner (A1)

Education

	Degree	Classification
2026/01 - 2030 Ongoing	Doutoramento em informática (Doctorate) Universidade de Lisboa Faculdade de Ciências, Portugal	
2021 - 2024/03/06 Concluded	Informática (Master's Degree) Universidade de Lisboa Faculdade de Ciências, Portugal ""Exploring How Robots can Help Older Adults Live by Their Values"" Final Classification: 18" (THESIS/DISSERTATION)	16
2020 - 2022 Concluded	Estatística Aplicada à Biologia e Ciências da Saúde (Post-Bachelor's Specialization) Universidade de Lisboa Faculdade de Ciências, Portugal	14.40
2017/09 - 2020/07 Concluded	Biologia (Licenciate Degree) Universidade de Lisboa Instituto Superior de Agronomia, Portugal ""Reproduction in captivity of Raja undulata"" (THESIS/DISSERTATION)	13

Affiliation

Science

	Category Host institution/organization	Employer
2024/11/06 - 2026/01	Researcher (Research) LASIGE Laboratório de Sistemas Informáticos de Grande Escala, Portugal	Universidade de Lisboa Faculdade de Ciências, Portugal
2024/02/22 - 2024/06/30	Researcher (Research) Kungliga Tekniska Högskolan Skolan för elektroteknik och datavetenskap, Sweden	Kungliga Tekniska Högskolan Skolan för elektroteknik och datavetenskap, Sweden
2020/02/17 - 2020/05/31	Research Trainee (Research) Aquário Vasco da Gama, Portugal	Aquário Vasco da Gama, Portugal

Others

	Category Host institution/organization	Employer
2016 - 2018	Tutor at a tutoring center. Provided individual and small-group educational support across various school subjects.	Instituto de Formação académica, Portugal

Projects

Grant

	Designation	Funders
2026/01/01 - 2029/12/31	Exploring Long-Term Trust in Human-Robot Interaction with Older Adults 2025.03576.BD 10.54499/2025.03576.BD	Fundação para a Ciência e a Tecnologia, Portugal Ongoing

Contract

	Designation	Funders
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2024/11/06 - 2025/11/04	HARIA: Human-Robot Sensorimotor Augmentation - Wearable Sensorimotor Interfaces and Supernumerary Robotic Limbs for Humans With Upper-Limb Disabilities HORIZON-CL4-2021-DIGITAL-EMERGING 01_101070292 Researcher FCiênciasID Associação para a Investigação e Desenvolvimento de Ciências, Portugal	Ongoing
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Other

	Designation	Funders
2025 - Current	EXPRESS-AR: Expressão Artística para pessoas com Tetraplegia usando Realidade Aumentada e Robótica - Researcher	Concluded

Outputs

Publications

Conference abstract	1	Nogueira, Beatriz; Simão, Hugo; Bernardino, Alexandre; Jodi Forlizzi; GUERREIRO, TIAGO. "Putting the Age in Agency: A Case Study of an Older Adult Controlling a Telepresence Robot in a Family Setting". Paper presented in <i>ACM Designing Interactive Systems Conference (DIS), Funchal, Madeira, 2025</i> . Submitted
Conference poster	1	<u>Nogueira, Beatriz; P. Ferreira, João</u> ; Simão, Hugo; Rocha, Filipa; Isabel Neto ; Guerreiro, João; GUERREIRO, TIAGO. Corresponding author: Nogueira, Beatriz. "Don't Go Breaking My Trust: A Taxonomy of Interaction Failures for Transparent Robots". Paper presented in <i>Designing Transparent and Understandable Robots (D-TUR) Workshop at HRI 2026, 2026</i> .
	2	<u>P. Ferreira, João; Nogueira, Beatriz</u> ; Simão, Hugo; Guerreiro, João; GUERREIRO, TIAGO. Corresponding author: P. Ferreira, João. "The Hidden Costs of Automation: De-Automation as a Design Principle for Human–Robot Interaction". Paper presented in <i>Equitable Robotics for Wellbeing (Eq-RW) workshop at HRI 2026, 2026</i> .
Journal article	1	Elmira Yadollahi; Fethiye Irmak Dogan; Yujing Zhang; Nogueira, Beatriz; GUERREIRO, TIAGO; Shelly Levy Tzedek; Iolanda Leite. "Expectations, Explanations, and Embodiment: Attempts at Robot Failure Recovery". <i>Journal of Human-Computer Studies</i> (2025): https://arxiv.org/pdf/2504.07266 . Under revision

Thesis /
Dissertation

1

"Exploring How Robots can Help Older Adults Live by Their Values". Master, Universidade de Lisboa Faculdade de Ciências, 2024. <http://hdl.handle.net/10451/64133>.

Activities

Event participation

	Activity description Type of event	Event name Institution / Organization
2024/10 - Current	Presentation at the department's Welcome Day, talking about the experience working at KTH and collaborating there on projects.	Welcome Day: Universidade de Lisboa LASIGE Laboratório de Sistemas Informáticos de Grande Escala, Portugal
2018 - Current	Graal ONGD "Jovens no Terraço" Participations. A series of talks regarding important subjects of our society, from equality, to sustainability and economy. Round table	"Jovens no Terraço" Graal - associação Social E Cultural, Portugal
2023/09/04 - 2023/09/08	AGORA young feminist summer school - European Women's Lobby Conference	AGORA young feminist summer school European Women's Lobby, Belgium
2023/03/31 - 2023/06/16	Parliament for the Future of Europe with the Portuguese Women's Platform DIGITAL TRANSFORMATION - 31 March 2023, Tallin, Estonia CLIMATE AND THE ENVIRONMENT - 16 June 2023, Burgas, Bulgaria Conference	Democracy International, Germany

Association member

	Society Organization	Role
2019/09 - 2020/06	Member of Biology Student Association at Instituto Superior de Agronomia (ISA), University of Lisboa	Organized academic events, lectures, and workshops.

Scientific expedition

	Activity description	Institution / Organization
2024/01/15 - 2024/06/30	Research Project: Advanced Adaptive Intelligent Systems The project aimed to develop adaptive social robots capable of understanding and responding to human communication and actions. The goal is	KTH- Royal School of Technology, Sweden

a seamless adaptation to users' needs and preferences with minimal interruption. A key focus was creating an intelligent kitchen assistant to aid in food preparation and kitchen tasks, particularly for supporting aging individuals. The robots will interact in real-time, track user emotions and actions, and adapt based on past interactions.

2024/01/15 - 2024/06/30	Research Project: Priming Robot Abilities Robots in human-collaborative tasks can fail and may need to ask for human help by explaining the failure. The explanation includes the cause and resolution. This research investigates how explanations can mediate people's expectations of robots. Participants were primed with positive or negative "priming videos" and then shown videos of robots experiencing communication failures, hardware failures, or social norm violations. In these scenarios, robots either explained or did not explain the failures.	KTH- Royal School of Technology, Sweden
2023/05/22 - 2023/05/27	Research visit at Carnegie Mellon University, in Pittsburgh to work on the ShiftHRI project, and discuss the thematic analysis of the acquired data.	Carnegie Mellon University, United States